

Volvo Trucks North America

Work Term Assignment for E S 225K

1st Term

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List of Acronyms

VTNA - Volvo Trucks North America

PSI - Propulsion Systems Integration

EATS - Exhaust Aftertreatment System

VAS - Volvo Autonomous Solutions

CARB - California Air Resources Board

EPA - Environmental Protection Agency

EMS - Engine Management System

ACM - Aftertreatment Control Module

TECU - Transmission Electronic Control Unit

EHCUC - Electronic Heater Control Unit

VMCU - Vehicle Main Control Unit

CAN - Controller Area Network

UQLS - Urea Quality Level Sensor

MIL - Malfunction Indicator Lamp

DPF - Diesel Particulate Filter

AHI - Aftertreatment Hydrocarbon Injector

I. Introduction

Founded in 1928, Volvo Trucks North America is a division of the Volvo Group serving the North American market. VTNA and Volvo Group subsidiary Mack Trucks, both of which I supported during my co-op, accounted for 16.8% of the U.S. medium and heavy-duty truck market in 2023 (Greenhalgh, 2024). The brand offers various truck models depending on use case (sleeper long-haul, day cab regional, vocational, etc) and powertrain (diesel, battery electric, and natural gas). VTNA and Mack Trucks are headquartered in Greensboro, North Carolina, with multiple other production facilities across the U.S. and connections to other Volvo Group facilities worldwide.

With over 19,600 employees across North America, VTNA focuses on high truck quality and performance, industry-leading safety, and environmental care (Volvo Group North America, n.d.). With these ideas in mind, VTNA seeks to have zero accidents involving Volvo trucks and 25% of the North American market share in medium and heavy-duty vehicles by 2030 (Welcome to Volvo Trucks, n.d.; Greenhalgh, 2024).

As a member of the Propulsion Systems Integration group, I was located at the Greensboro, NC location, in a smaller team away from the main Volvo Powertrain facility, which is located in Hagerstown, MD. The PSI group is split into the Vendor team, the Production Vehicle Evaluation team, and the Truck Support team, respectively. Specifically, as a Powertrain Integration co-op, I worked on the Truck Support team with 3 other engineers, assisting in integrating both Volvo and Mack powertrain systems with the chassis and complete vehicle groups, troubleshooting powertrain-related faults, and conducting performance checks and sign-offs on powertrain hardware and software updates. I was the only co-op on my team and one of 10 total interns at the Greensboro headquarters for the semester.

II. Work Environment

At Volvo, I worked closely with the Truck Support team members, including my primary mentor, Ronald Nobles, and engineers Kevin Stein and Krishna Vamsee Duggaraju. Ron is a senior engineer serving as the Truck Support team lead, focused on the integration and validation testing for all active projects, as well as maintaining project and troubleshooting documentation. He has a Bachelor of Science in both electrical and mechanical engineering, serving with the company since 1998, working in multiple departments, such as the powertrain software group, EATS group, and now PSI. Kevin is a senior engineer on the Truck Support team, focusing primarily on integration and validation testing for the VAS and second-generation VNL projects. He has a Bachelor of Science in mechanical engineering and has specialized in diesel engineering for 10 years, previously at Navistar and New Holland, as well as 10 years in the power generation industry as a controls field engineer at General Electric. Krishna is a senior engineer on the Truck Support team, focusing primarily on the integration and validation testing for upcoming CARB and EPA emissions projects. He has a Bachelor's and Master's of Science in mechanical engineering, specializing in controls engineering, with previous roles at Scania Trucks India and Pfizer.

As a part of the Truck Support team, I attended daily morning update meetings where we each described the status of projects we were working on, any roadblocks we faced, as well as any project info that may help in troubleshooting other projects. Additionally, I attended project specific meetings as needed, to stay up to date on the timelines and deliverables for the current projects I was assisting with (EPA and second-generation most notably), as well as a weekly update meeting across the entire PSI team, which consisted of 9 on-site engineers and 5 off-site engineers at the Hagerstown powertrain facility.

As part of the Volvo co-op program, I presented my contribution to the team's 2025 Strategic Objectives at the midpoint and end of the program, showcasing the Python data analysis tools I self-started to achieve one of our objectives. From these presentations, I garnered feedback on how to improve my tools and continue growing as an engineer.

III. Technical Summary

Truck Support and Root Cause Analysis

As part of the Truck Support team, I was primarily tasked with assisting in the integration and validation testing of all current projects, troubleshooting trucks, and performing root cause analysis, with the goal of resolving integration issues as they arose. This meant that I did not necessarily have a single project assigned to me as an intern, but rather, I was given assignments and projects as needed, varying in difficulty and span, depending on the issues that needed validation testing at the time. The projects I focused the most on included second-generation, VAS, CARB, and EPA models, all being truck variants designed for various applications and emissions regulations. Common issues and challenges that required troubleshooting from the Truck Support team included faulty powertrain hardware (clogged fuel lines, pneumatic gear selection system pressure loss, etc), electrical malfunctions (incorrect wire continuity, impedance resistance values being too high/low, etc), as well as engine control software bugs as software packages were developed and updated.

To perform validation testing and begin troubleshooting each truck, I first learned hands-on skills, such as reading faults and diagnostic trouble codes (see Figure 1) across powertrain and vehicle electronic control units, including the EMS, ACM, TECU, EHCUC, and VMCU. I also utilized data acquisition software such as ATI Vision and Vector CANalyzer to record sensor parameters and signal values during truck operation (see Figure 2).

Find :

Next

		Node	Status	Status I	Identifier	DTC	Caption	FTB	FTB Caption
		0x61	Active	0xAF	D1D21	U101900 (0xd01900)	Lost Communication With Services And Entertainment Moc	0x00	No Sub Type Information
		0x24	Active	0xAF	D1F0U	B105F87 (0x905f87)	Power Distribution Component for Instrument Panel Equipr	0x87	Missing Message
		0x40	Active	0x29	D1BUL	B151095 (0x951095)	Flexible Switch	0x95	Incorrect Assembly
		0x10	Active	0xED	D1B55	P019064 (0x019064)	Fuel Rail Pressure Sensor Bank 1	0x64	Signal Plausibility Failure
		0xA2	Inactive	0x28	D1A6F	U003788 (0xc03788)	CAN Communication Cab Net	0x88	Bus off
		0x1A	Inactive	0x20	D1AD0	U300049 (0xf00049)	Electronic control unit	0x49	Internal Electronic Failure
		0x51	Inactive	0x20	D1A76	U100387 (0xd00387)	Lost Communication With DACU	0x87	Missing Message

Figure 1: Example of Fault Code Reading

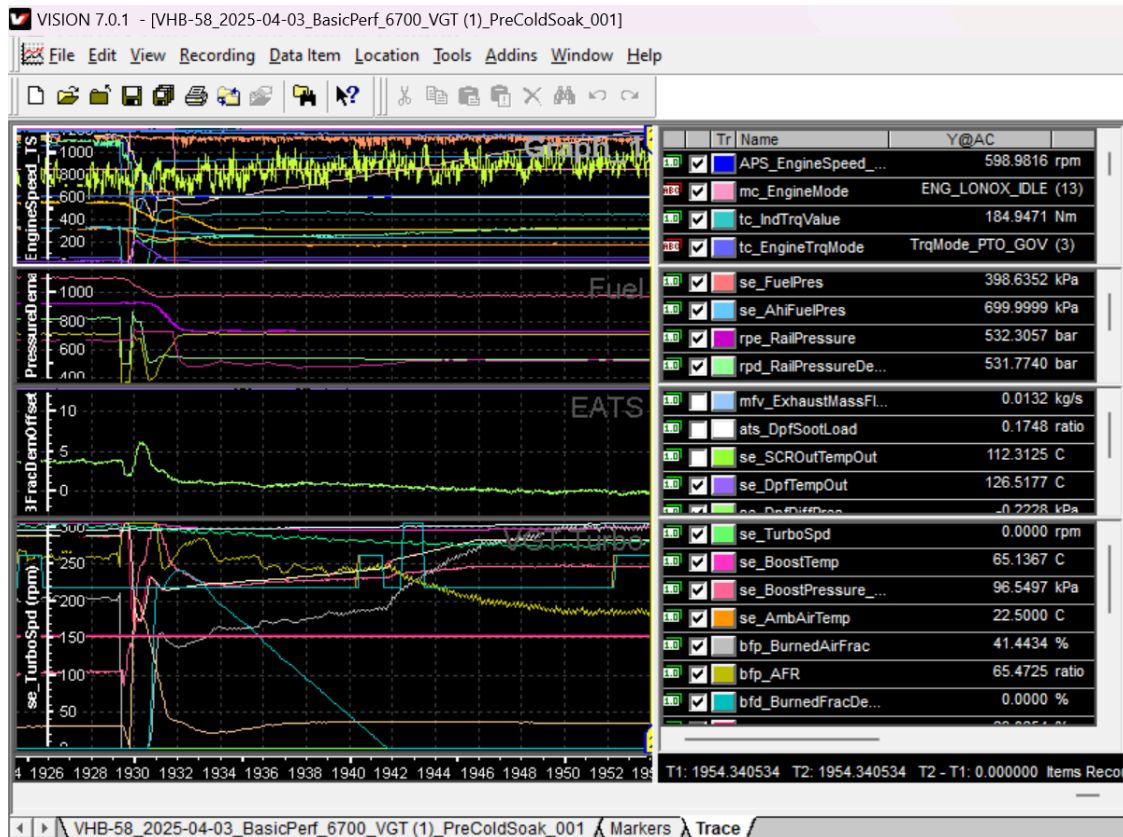


Figure 2: Example of ATI Vision Data Recording Screen

Furthermore, I also became comfortable using common electrical tools like standard and clamp multimeters to validate electrical wiring harnesses and circuits. Additionally, I was given access to function documentation and wiring diagrams, showcasing the control logic and pinouts of each truck's sensors and electronic components, which I used along with data analysis to find the root cause of powertrain faults and ensure proper system integration.

For example, in collaboration with Krishna on a prototype EPA build, our goal was to validate the truck's powertrain functions after new hardware and software installation, before road approval and further vehicle testing. Due to the nature of a prototype build, faults were present in the powertrain system during the initial integration, with one specific fault referring to lost communication between a sensor and its designated control units (EMS to ACM) during the key-on and engine running modes. To start root cause analysis, we first read the faults produced from the truck, referencing the function documentation and wiring harness pinouts to understand the possible locations and causes of each fault. Knowing we had a lost communication fault, we first started with software verification, ensuring that both units could properly attempt connection to our data acquisition systems independently, which would show that the software installation was successful, allowing data transfer, eliminating software as the culprit. After noticing the EMS properly synchronized with our software, while the ACM showed offline with no physical connection detected, we shifted our focus to a potential hardware issue by sectioning off the wiring harnesses between the EMS and ACM, checking continuity, voltage, and resistance to validate connectivity. Although we confirmed continuity between the two control units, a discrepancy in resistance values was identified, which exceeded the harness's specified resistance value. Because of this, the impedance resistance that CAN message transmission relies on was a different value, leading to lost communication. To resolve the issue, we performed a point-to-point circuit inspection, checking for any crossed CAN high and CAN low wiring that could lead to irregular voltage differential readings, and repopulating electrical pins to their correct positions if needed. Ultimately, the issue was caused by a faulty UQLS module in the circuit, which caused the irregular resistance values. Our troubleshooting cleared the fault, and the truck became one step closer to proper system integration with the rest of the vehicle.

Additionally, in collaboration with Kevin on an updated second-generation VNL truck, we had been issued a work ticket regarding a low coolant temperature fault on a field test truck that had triggered the MIL on the instrument cluster. Our objective was to restore the coolant system functionality and clear the fault. To start root cause analysis, we devised a test plan to gather any sensor data related to the coolant system while trying to reproduce the fault. This included over 25 engine parameters, such as coolant temperature, engine mode, and the engine warmup completed boolean flag, which had caused the active fault. From there, I helped set up data logging through ATI Vision and Vector CANalyzer on the truck's powertrain CAN bus, where we then started the truck, placing a high RPM load on the truck's engine to elevate coolant temperatures. Unable to initially reproduce the fault, we contacted component owners of the coolant system, and I referred to the function documentation of the coolant temperature sensor and its associated warmup flag. Because of this, we were able to dissect the logic behind the production of the fault and better understand the fault's triggers, which consisted of idle time, ambient temperature, and coolant temperature. Over the next week, with a revised test plan and new understanding of the warmup flag functionality, tests were conducted with the truck running at idle, high engine load while parked, and even in a loop on the facility's test track, to collect valuable data under multiple scenarios, further understanding the fault and how to turn the warmup completed flag to true. Through this, we were able to get the fault to reproduce and ultimately become cleared while also gaining a better understanding of the coolant system's functionality and integration with the vehicle's other control units, which proved to be a useful tool in future coolant system troubleshooting. These two examples show the general process of troubleshooting trucks across different projects, as well as the thought process and tools my team and I employed to help ensure proper system functionality and integration on each truck.

Ultimately, this helped us with our goal of keeping trucks running and learning about potential failure points on the powertrain.

Active Regenerations

Another smaller recurring task of mine was to conduct parked EATS active regenerations (regens) on trucks before the sign-off and eventual road approval of powertrain systems. The goal of an active regeneration on a diesel truck is to clean any soot buildup from truck operation within the EATS (example configuration shown in Figure 3), mainly the DPF, through the use of controlled hydrocarbon injection and high temperatures.

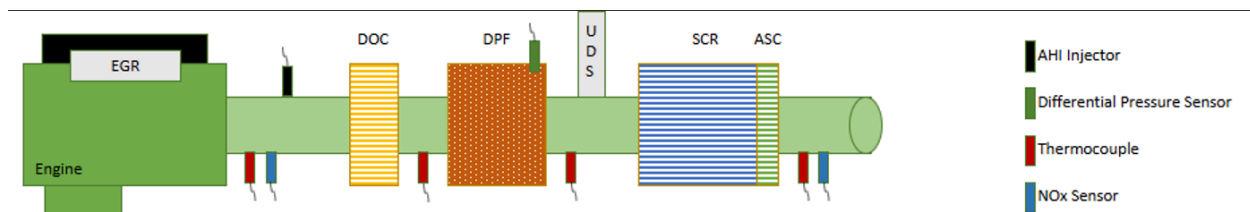


Figure 3: Example Layout of EATS

This process entails flashing an engine calibration with forced high soot ratio values onto the engine by connecting to the truck's CAN bus with ATI Vision software, which prompts a regen. Due to the engine load and full engagement of the exhaust aftertreatment system during the regen process, as well as the importance of proper regen performance for vehicle reliability and longevity, the goal of using this process is to act as a benchmark for powertrain functionality. Upon starting the regen, I would monitor sensor and parameter data through the ATI Vision software, such as exhaust temperatures and AHI fuel pressure. Additionally, I would visually inspect the engine and exhaust components, checking around the engine's air intake and exhaust clamps for air leaks, any visible fuel, oil, or coolant leaks, as well as any abnormal noises that could signal issues like faulty engine timing or heavy vibrations during the regen. After the successful completion of each regen, along with verifying the condition of the powertrain

hardware, I collected relevant vehicle information, such as fuel injector trim codes, engine ID, and active engine calibration. Using the collected data and diagnostics, I prepared the truck for sign-off and documented a verified timestamp of successful powertrain operation.

Python Data Analysis

Along with these tasks, I also self-taught myself Python to create successful and impactful programs for our yearly strategic objectives. Specifically, I focused on implementing Python code that created and managed a historical database of powertrain data. I created code where older historical data could be easily referred to and plotted on figures and charts with current data to initiate quicker reactive troubleshooting and help proactively point out potential issues before other engineers or customers find them in the field. I was inspired to create these tools when my team continued to mention that the mass amounts of data taken while troubleshooting problems daily were sometimes lost as new problems arose and projects shifted, meaning that problems we had seen and solved before could not be easily applied to repeat fault occurrences, which hurt our team's efficiency. Not to mention, my team's current software tools for comparison between multiple trucks were inefficient and hard to use, making it harder to visualize consistency between trucks as well as changes in parameter and sensor values across projects. During downtime between Truck Support projects, I created Python scripts, utilizing the Pandas and Numpy libraries for data processing and manipulation, the Matplotlib library for data visualization, the PyPDF and FPDF libraries for PDF generation, as well as the Tkinter library for GUI creation. Combined with other built-in functions, I coded a user-configured graph generator with stored databases .pkl files that dynamically detect comparable parameters and change axis properties based on the properties of each input data file. The graph figure also features a live cursor readout of plotted values and is saved to a PDF output file each time the

script is run. Additionally, with the help of my team, I scaled the project to be increasingly user-friendly by receiving feedback on the tool's application and usability in our troubleshooting ecosystem, adapting the tool over time.

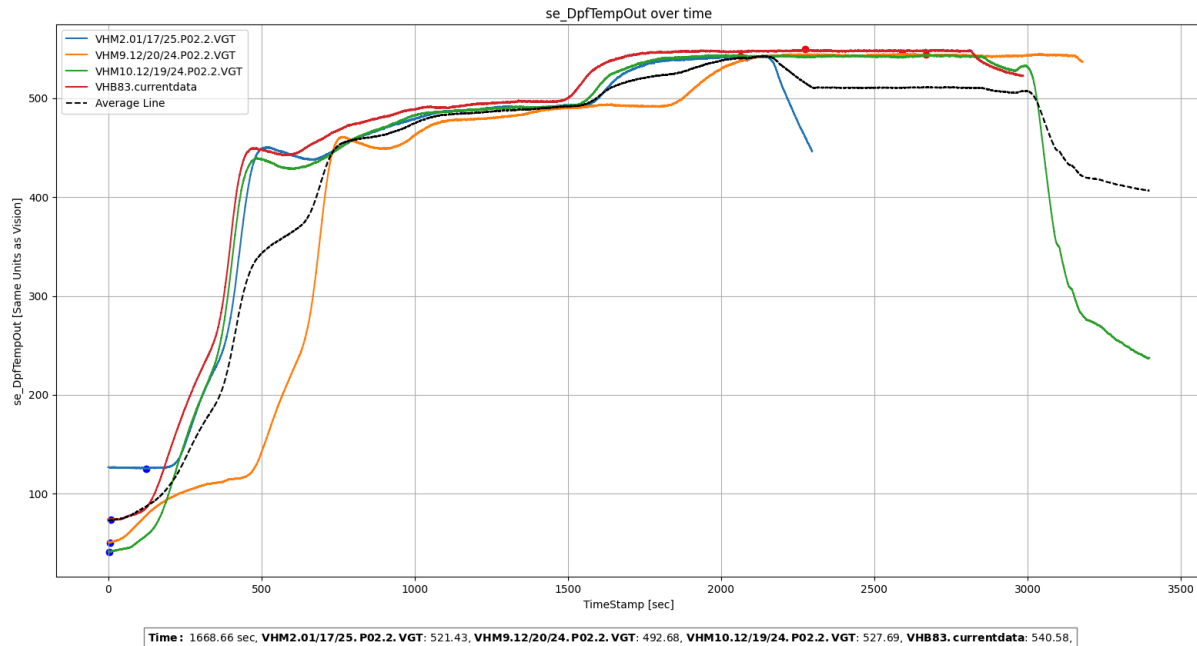


Figure 4: Generated Graph for DPF Temperature Over Time Across Multiple Trucks

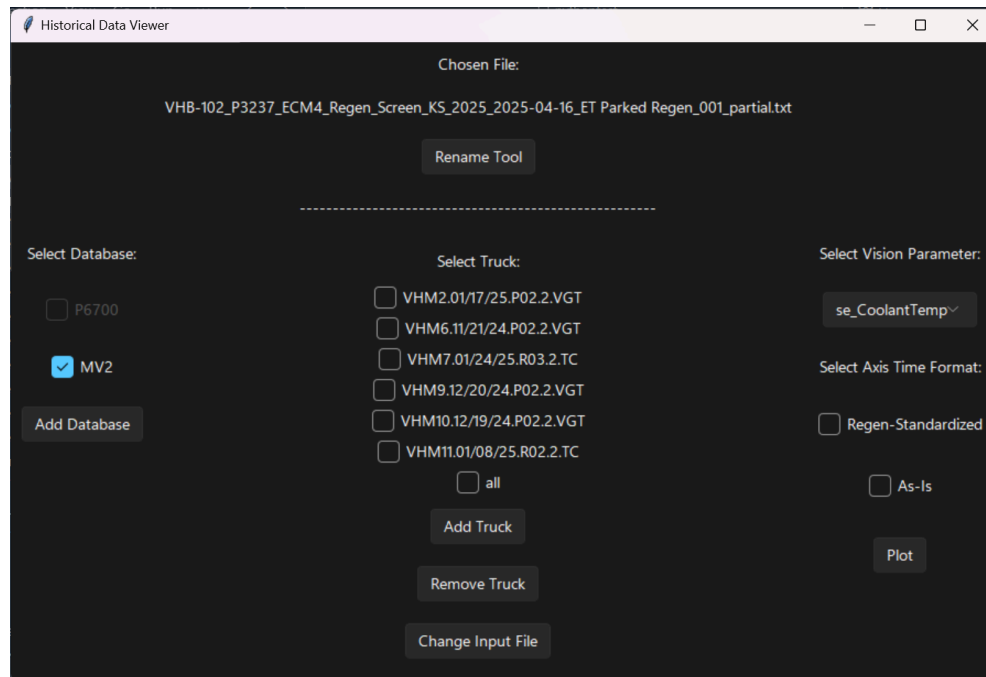


Figure 5: Script UI Created with Tkinter Library

IV. Term In Review

My time with Volvo Group has given me a great first look at the automotive industry. My prior knowledge of vehicle and overall engineering design from student organizations like Longhorn Racing, as well as my work on my personal vehicle projects, I feel, gave me a great starting foundation for this opportunity. In terms of my professional skills, I feel I have strengthened my adaptability and troubleshooting skills, with the ability to quickly change gears between projects, designs, and systems to effectively solve problems. Looking ahead to my next co-op term, I want to further challenge myself by embracing unfamiliar systems and asking more questions without hesitation. Through my early confusion in this co-op position, I recognized that growth often lies in the areas I understand the least, which may lead to new interests and even opportunities. By far, the most valuable experience of this co-op opportunity was the exposure to the working processes and innovative technologies of a large automotive company, with the least valuable being the sometimes inconsistent projects and workload, a natural outcome of being at a large automotive company. I plan to use my new skills in powertrain engineering and data acquisition to keep exploring my passion in the automotive industry, potentially enrolling in related CGE courses such as Combustion Engine Processes and Data Science for Engineers, as well as continuing to work with other students on automotive related projects like Formula SAE, with a focus on engine and powertrain development.

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